

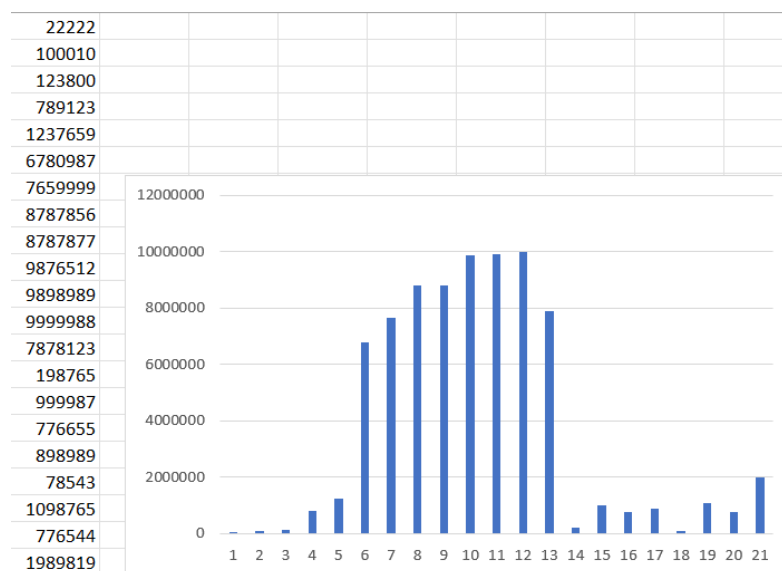
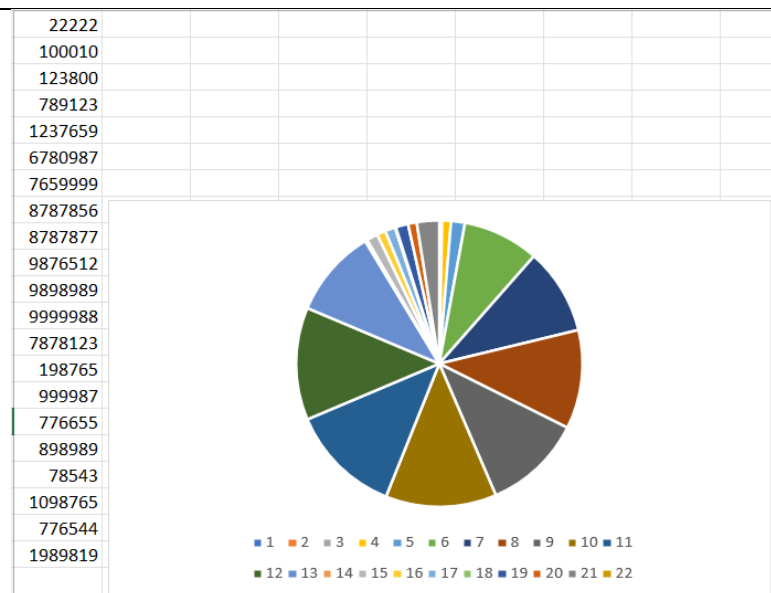
AP Computer Science Principles
Ticket Out the Door
Set 1: Exploring One-Column Data

Name _____ Period _____

Skill 1.1 Exercise 1

For each data set below, identify the largest number and the smallest number.

100010	636256	9898989	12121212	789123	123800	8907618
9875671	10101010	98111111	8787877	98989891		
9876512	6780987	1237659	89898910	7424048	76654321	
8787856	7659999	99887766	78956430	12914651		
9999988	1628760	2222290	10926382	9187656	10987654	



Name _____ Period _____

Skill 1.1 Exercise 2

For which data set was it easiest to identify the largest and smallest number? Why?

Why do people make visualizations out of data?

Skill 1.2 Exercise 1

List the steps in the data analysis process.

Skill 1.3 Exercise 1

Below is a portion of a data set that contains information about different dog breeds. For each column, indicate whether it stores quantitative or qualitative data.

Breed Name	Breed Group	Bred For	Min Life	Max	Min Height	Max Height	Min Weight	Max Weight	Temperament
Affenpinscher	Toy	Small rodent hunting, lapdog	10	12	9	12	6	13	Stubborn, Curious, Playful, Adventurous, Active, Fun-loving
Afghan Hound	Hound	Coursing and hunting	10	13	25	27	50	60	Aloof, Clownish, Dignified, Independent, Happy
Airedale Terrier	Terrier	Badger, otter hunting	10	13	21	23	40	65	Outgoing, Friendly, Alert, Confident, Intelligent, Courageous

Column	Data type (quantitative/qualitative)
Breed Name	
Breed Group	
Bred For	
Min Life	
Max Life	

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Name _____ Period _____

Min Height		
Max Height		
Min Weight		
Max Weight		
Temperment		

Skill 1.4 Exercise 1

Open the dog data set, <https://docs.google.com/spreadsheets/d/1dy2TrqRqXNcq-0k4ciLcATcPINv8u1eNPWhbJRuYYjU/edit?usp=sharing>

For each column indicate whether the data would be best visualized using a bar chart, histogram chart, or neither.

Column	Histogram, Bar, or Neither
Name	
Breed Group	
Bred For	
Minimum Life Span	
Maximum Life Span	
Minimum Height	
Maximum Height	
Minimum Weight	
Maximum Weight	
Temperament	
Image	

Name _____ Period _____

For each column you indicated as best being visualized as a *histogram*, indicate a corresponding bucket size that would be suitable for visualizing the data.

Column	Bucket Size

Skill 1.5 Exercise 1

The data below was collected from students using a survey. Indicate how you would *clean* the data by crossing out the incorrectly formatted data and replacing it with what was intended.

id	Age	Grade	FavoriteSubject	AverageHoursOfSleep	AverageHoursOfEntertainment
1		16	10 Math	7	2
2		15	10 Spanish	6	3
3		16	11 Spanish	7	two
4		17	10 CS	8	2
5		15	9 History	six	one
6		18	12 Computer Science	6	3
7		17	11 Bio	5	3
8		15	9 English	5	3
9		18	11 Music	8	0
10		15	11 Computer Science	8	3
11		15	11 Art	7	0
12		18	9 Computer Science	6	4
13	sixteen	nine	Art	6	1
14		15	9 Computer Science	6	0

You are a developer and have been asked to create a web-based form to collect the above data. What could you do to better ensure the user entered clean data?